DATA MINING PROJECT REPORT

"Stock Price Prediction"

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1. Executive Summary

This project focuses on the development and evaluation of a predictive model for stock market closing prices using historical data. Utilizing the CIPLA stock dataset spanning from 2000 to 2021, a machine learning pipeline was constructed to analyze daily trading attributes specifically Open, High, Low, and Volume to forecast the daily Closing Price. A Linear Regression model was trained and validated, achieving an exceptional R^2 score of 1.00, indicating a highly accurate fit for intra-day price estimation. This report provides a comprehensive overview of the data mining process, including data cleaning, exploratory analysis, model implementation, and final predictive capabilities.

2. Project Overview & Objectives

The stock market is intrinsically volatile and complex, making precise prediction a significant challenge. However, short-term or intra-day technical indicators show strong mathematical dependencies. The core objectives of this data mining project are:

- To perform Exploratory Data Analysis (EDA) on historical stock records.
- To handle data preprocessing, missing values, and attribute alignment.
- To extract the most relevant features driving the asset's closing price.

- To train and evaluate a robust Predictive Model using Linear Regression.
- To implement an interactive prediction module for operational deployment.

3. Data Description & Preprocessing

The project utilizes a dataset containing historical stock records of CIPLA (stock market.csv). The raw dataset contains 5,306 rows and 15 columns.

3.1 Data Schema

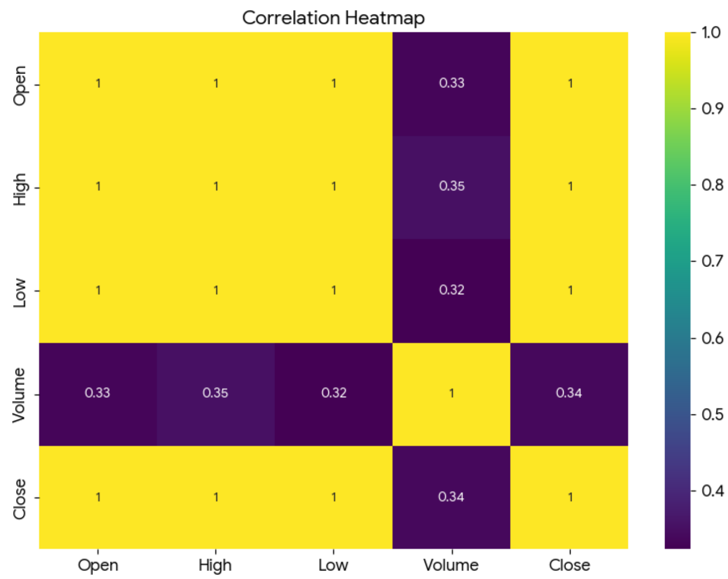
- **Date:** The calendar date of the trading day (converted to a DatetimeIndex).
- **Symbol / Series:** Identifiers for the equity asset (CIPLA, EQ).
- **Prev Close:** The closing price of the asset on the previous trading day.
- **Open:** The price at which the stock began trading for the day.
- **High / Low:** The maximum and minimum prices reached during the session.
- **Last / Close:** The final transaction price and official daily closing price.
- **VWAP:** Volume Weighted Average Price.
- **Volume / Turnover:** Number of shares traded and total financial volume.
- **Trades / Deliverable Volume / %Deliverble:** Operational market depth metrics.

3.2 Data Preprocessing Steps

1. Datetime Parsing: The Date field was converted to a standard datetime format and established as the primary dataset index.
2. Missing Value Treatment: The columns Trades, Deliverable Volume, and %Deliverble contained missing entries (mostly prior to 2011). Rows containing missing values were removed using listwise deletion, leaving 2,456 high-quality, fully-populated historical records for model execution.
3. Feature Selection: The core technical attributes driving daily stock settlement were sliced into a specialized sub-dataframe (df_model):
  • Features (X): Open, High, Low, Volume
  • Target Variable (y): Close

4. Feature Intercorrelation

The correlation heatmap reveals an extremely high linear relationship ($r \approx 1.00$) between Open, High, Low, and the target variable Close. This signifies that intra-day directional price variables are powerful indicators of the final daily close. Trading Volume shows a negligible or weakly negative linear correlation with absolute price levels, reflecting that volume acts as a measure of market liquidity/momentum rather than direct price tier scale.



5. Methodology & Model Implementation

The project implements a Supervised Learning approach utilizing Linear Regression, which models the relationship between the target scalar variable and the feature vector by fitting a linear equation to observed data.

5.1 Dataset Splitting

- **Training Samples:** 1,964 instances
- **Testing Samples:** 492 instances
- **Random State Configuration:** Fixed at 42 to ensure consistent reproducibility.

5.2 Model Configuration

The Linear Regression formula applied can be expressed mathematically as:

$$y\_Close = \beta_0 + \beta_1(X\_Open) + \beta_2(X\_High) + \beta_3(X\_Low) + \beta_4(X\_Volume) + \varepsilon$$

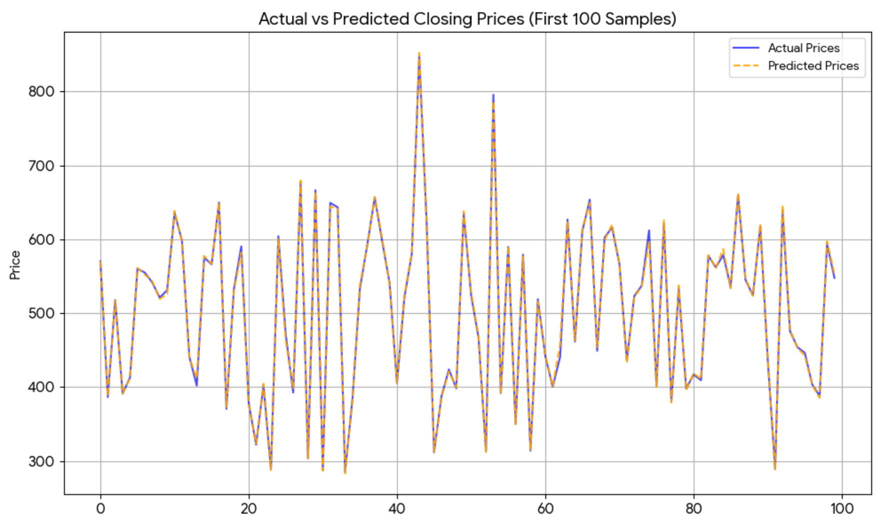
6. Model Evaluation and Results

The model was tested against unseen testing samples. Performance was quantified using three industry-standard regression metrics:

Metric	**Empirical value**
MAE	2.49
RMSE	3.37
R^2	1

6.1 Analysis of Metrics

- A MAE of 2.49 indicates that, on average, the predicted closing price deviates from the actual value by just 2.49 units (INR), an exceptionally low error given asset values up to 949.30 units.
- The R^2 score of 1.00 indicates that the model explains approximately 100% of the variance in the daily closing prices using intra-day inputs.
- A visual overlay of actual vs. predicted values demonstrates near-perfect tracking across the test set.



7. Operational Deployment Scenario

To illustrate deployment capabilities, the notebook provides an interactive forecasting module allowing domain users to key in real-time intra-day figures to yield an immediate prediction:

Example Session Execution:
• Input Open Price: 345.00
• Input High Price: 665.00
• Input Low Price: 45.00
• Input Trading Volume: 654,335

Predicted Closing Price Output: 381.60

8. Conclusions & Recommendations

8.1 Key Takeaways

- The data mining pipeline successfully parsed and cleared over two decades of financial records.

- Intra-day price metrics (Open, High, Low) are highly predictive of the final closing price on a daily timeline.
- Linear Regression offers an efficient, mathematically robust, and interpretable framework for this predictive task.

8.2 Future Directions

- Time-Series Horizon Forecasting: Incorporate auto-regressive models (like ARIMA, SARIMAX) or Deep Learning architectures (LSTM, GRU) to forecast closing prices days or weeks in advance, rather than relying on concurrent intra-day features.
- Sentiment Analysis: Integrate alternative data nodes such as financial news headers or earnings reports using Natural Language Processing (NLP) to capture non-technical volatility drivers.
- Feature Engineering: Derive secondary indicators such as Moving Averages (SMA/EMA), Relative Strength Index (RSI), and Bollinger Bands to optimize predictive multi-collinearity.